

# Bolle in un superfluido ferromagnetico

Giorgio Micaglio

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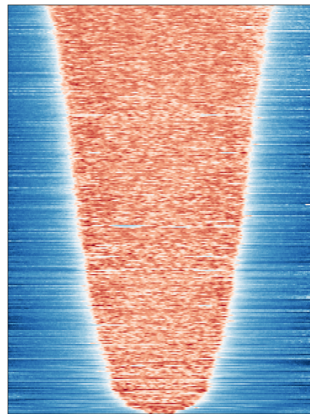


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Questa presentazione ricoprirà:

- ▶ **Introduzione**
- ▶ **Background teorico:** Ferromagnetismo in miscele di spin a due componenti
- ▶ **Analisi dati:** Caratterizzazione di bolle di decadimento di falso vuoto
- ▶ **Conclusioni**



## Perché bolle in un superfluido ferromagnetico?

- ▶ Estrema figaggine del titolo
- ▶ Prima **osservazione sperimentale** del decadimento da falso vuoto (false vacuum decay, FVD) tramite formazione di bolle nei laboratori del *Pitaevskii BEC Center* dell'Università di Trento.
- ▶ **Superfluidità**: alto grado di coerenza del sistema, singola funzione d'onda
- ▶ **Ferromagnetismo**: panorama energetico a doppia buca
- ▶ Lo studio del FVD dà informazioni sulla metastabilità, dai sistemi quantistici alla cosmologia

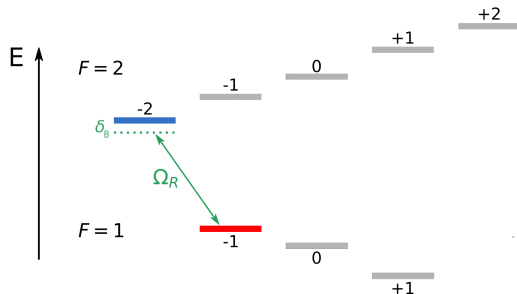
# Piattaforma sperimentale

- $N \sim 10^6$  atomi di  $^{23}\text{Na}$  (condensati) negli stati iperfini:

$$|F = 2, m_F = -2\rangle = |\uparrow\rangle$$

$$|F = 1, m_F = -1\rangle = |\downarrow\rangle$$

- Costanti di interazione di contatto  $g_{\uparrow\uparrow}$ ,  $g_{\downarrow\downarrow}$ ,  $g_{\uparrow\downarrow}$
- Interconversione possibile per accoppiamento Rabi: strength  $\Omega_R$  e detuning  $\delta_B$
- Potenziale di intrappolamento armonico (forma di sigaro, 1D)





# Miscele di spin a due componenti accoppiate

Le GPE contengono le costanti di interazione **intra** e **inter**-componente interaction constants e l'**accoppiamento** tra i due stati:

$$\begin{aligned} \left[ -\frac{\hbar^2}{2m} \nabla^2 + V(x) - \frac{\delta_B}{2} + g_{\uparrow\uparrow} |\psi_{\uparrow}(x)|^2 + g_{\uparrow\downarrow} |\psi_{\downarrow}(x)|^2 \right] \psi_{\uparrow}(x) - \frac{\hbar\Omega_R}{2} \psi_{\downarrow}(x) &= \mu_{\uparrow} \psi_{\uparrow}(x) \\ \left[ -\frac{\hbar^2}{2m} \nabla^2 + V(x) + \frac{\delta_B}{2} + g_{\downarrow\uparrow} |\psi_{\uparrow}(x)|^2 + g_{\downarrow\downarrow} |\psi_{\downarrow}(x)|^2 \right] \psi_{\downarrow}(x) - \frac{\hbar\Omega_R}{2} \psi_{\uparrow}(x) &= \mu_{\downarrow} \psi_{\downarrow}(x) \end{aligned}$$

# Miscele di spin a due componenti accoppiate

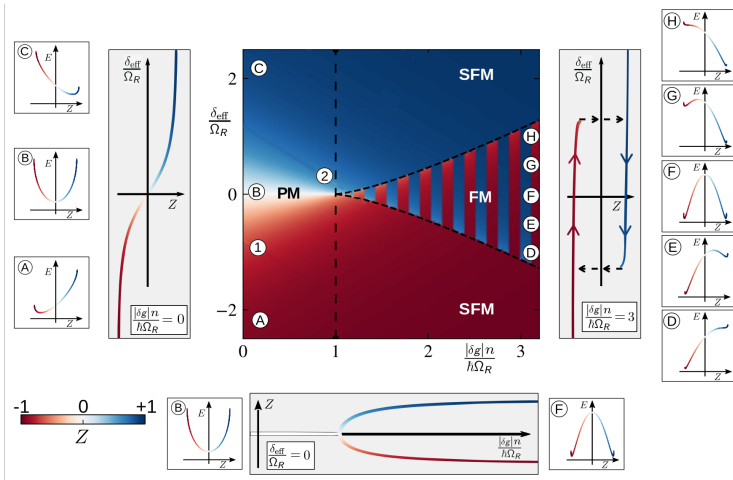
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La magnetizzazione  $Z = (n_{\uparrow} - n_{\downarrow}) / (n_{\uparrow} + n_{\downarrow})$  produce un panorama energetico a doppia buca:

$$E_{\text{MF}}(Z) = -|\delta g| n Z^2 - 2\hbar\Omega_R \sqrt{1 - Z^2} - 2\hbar\delta_{\text{eff}} Z$$

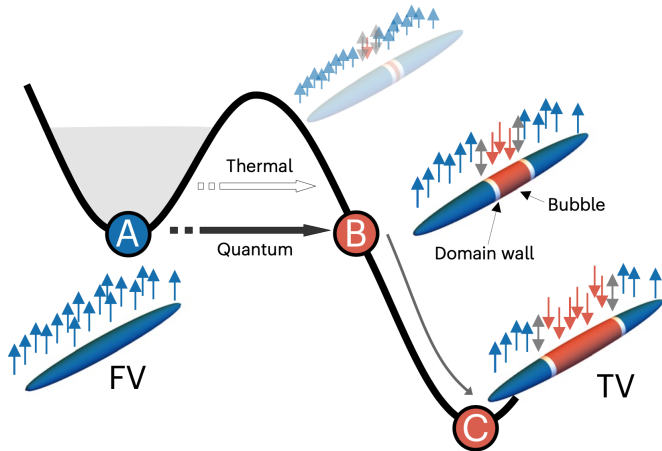
# Diagramma di fase del modello magnetico: configurazione del GS



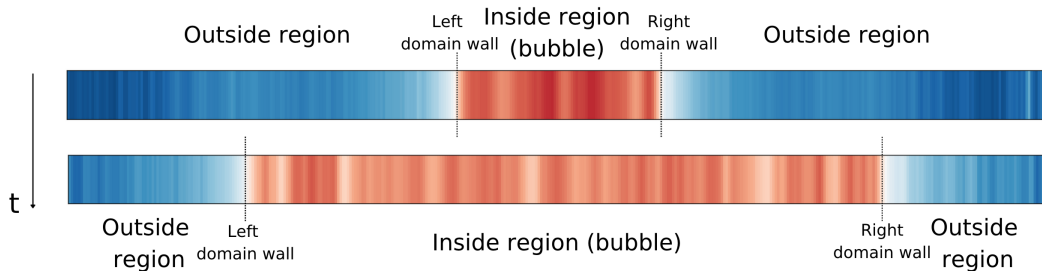
- Fisso  $\frac{|\delta g|n}{\hbar \Omega_R}$  ad un valore fissato cambiando  $\Omega_R$
- Sondo le proprietà magnetiche cambiando  $\delta_{\text{eff}}$

# Bolle di false vacuum decay

- Effetto tunnel quantistico da A to B (stocastico)
- Decadimento da B a C (ciò che vogliamo studiare)
- Problema: quando possiamo fotografare il sistema?



# Esempio di foto di una bolla

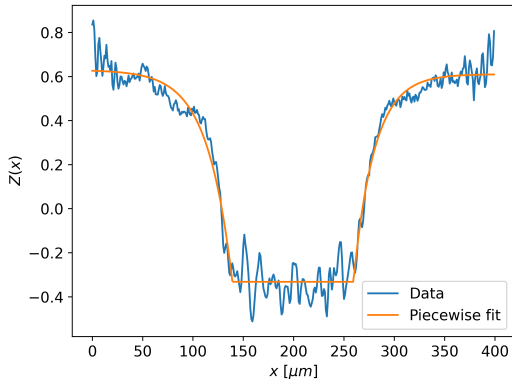
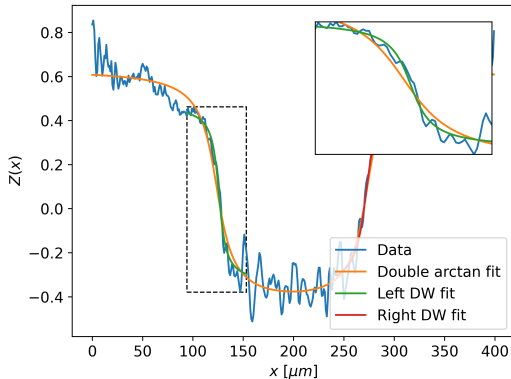


Il nostro obiettivo è studiare:

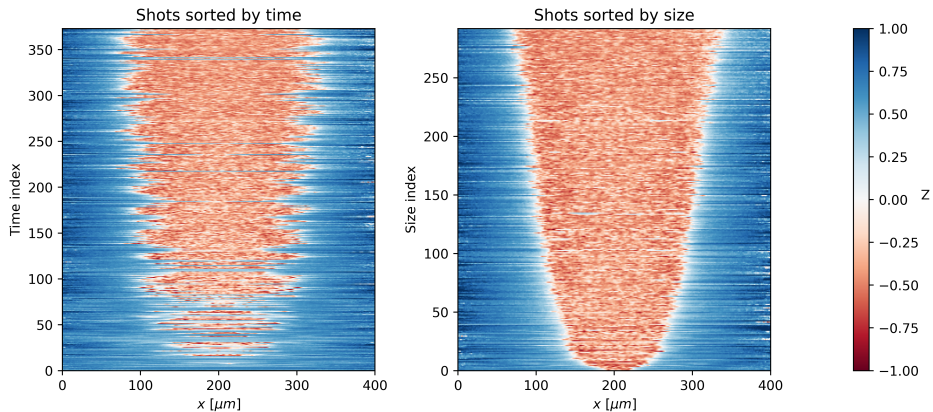
- ▶ La dimensione  $\sigma_B$  e la larghezza dei domain wall  $w_D$
- ▶ Strutture di eccitazione nelle regioni interno/esterno

# Bubble fitting routines

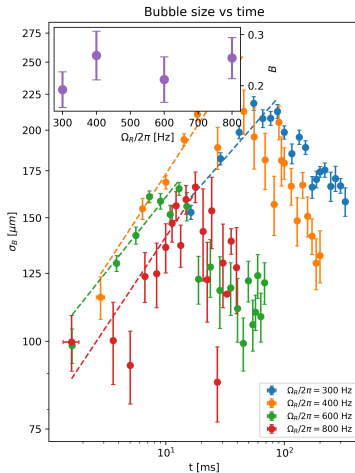
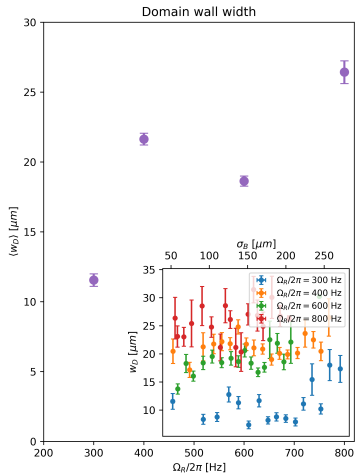
Example of fitting routines,  $\Omega_R/2\pi = 400$  Hz and  $\delta = 596.5$  Hz



Bubble shots with  $\Omega_R/2\pi = 400$  Hz and  $\delta = 596.5$  Hz



# Analysis of domain wall width and bubble size



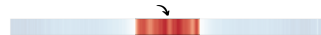
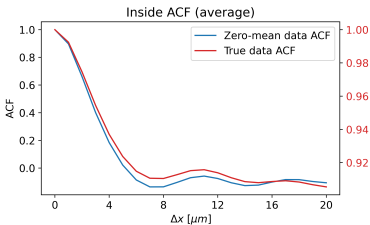
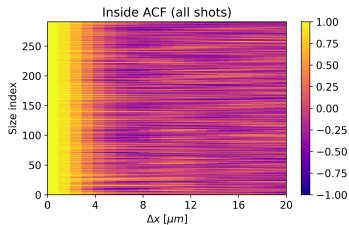
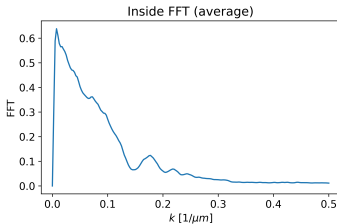
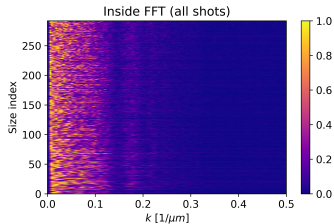
- Shots clustered by size (left) or time (right)
- Fit function:

$$\sigma_B(t) = A \left( \frac{t}{1 \text{ ms}} \right)^B$$



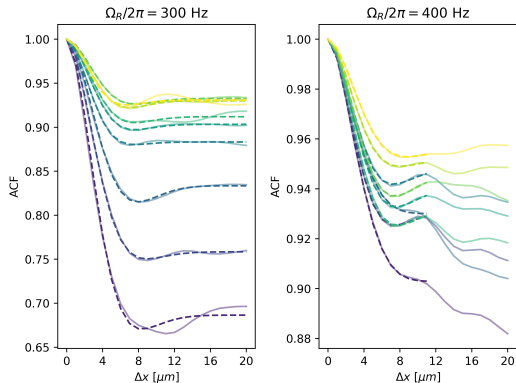
# Spectral analysis in the inside region

FFT and ACF on shots with  $\Omega_R/2\pi = 400$  Hz and  $\delta = 596.5$  Hz



- ▶ FFT: broad peak at  $k \sim 0.01 \mu\text{m}^{-1}$
- ▶ ACF: peak at  $\Delta x \sim 11 \mu\text{m}$

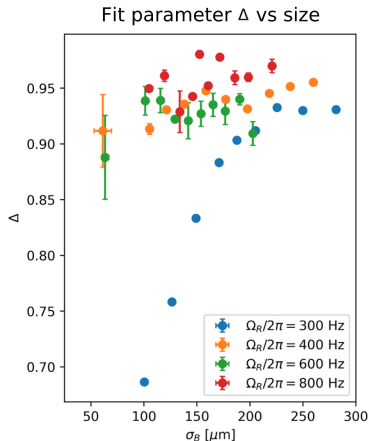
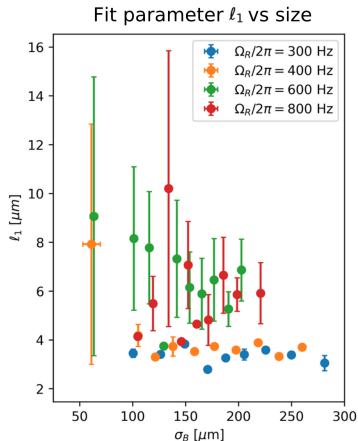
# ACF analysis in the inside region vs $\sigma_B$



- ▶ Shots clustered by increasing size (darker to lighter colors)
- ▶ Fit function  $\mathcal{A}_{\text{fit}}(x) =$

$$(1-\Delta) \cos\left(\frac{\pi x}{\ell_2}\right) \exp\left[-\frac{1}{2} \left(\frac{x}{\ell_1}\right)^{1.7}\right] + \Delta$$

# ACF analysis in the inside region vs $\sigma_B$



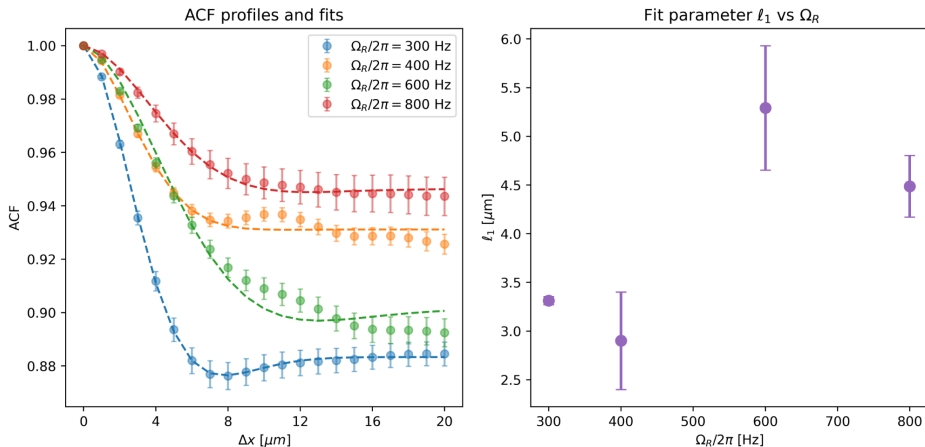
Healing lengths:

$$\xi_s = \frac{\hbar}{\sqrt{2mn|\delta g|}} \approx 0.5 \mu\text{m}$$

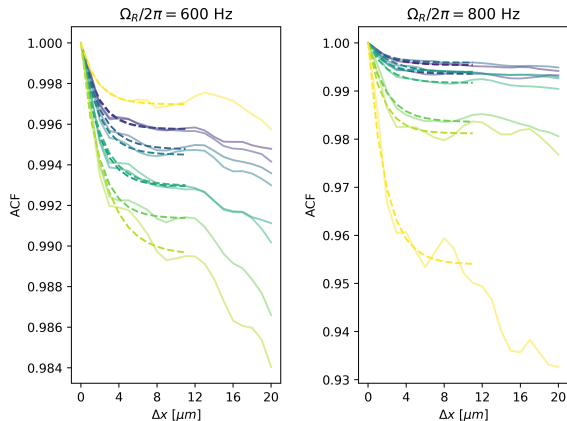
$$\xi_d = \frac{\hbar}{\sqrt{2mn\bar{g}}} \approx 0.01 \mu\text{m}$$

$$\xi_R = \sqrt{\frac{\hbar}{m\Omega_R}} \approx 1.8 - 3.0 \mu\text{m}$$

# ACF analysis in the inside region vs $\Omega_R$



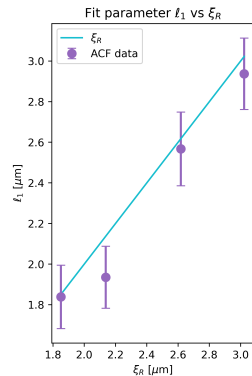
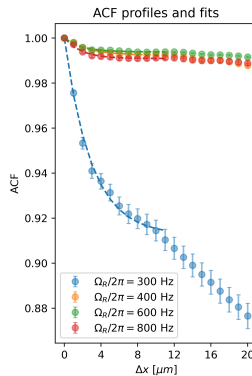
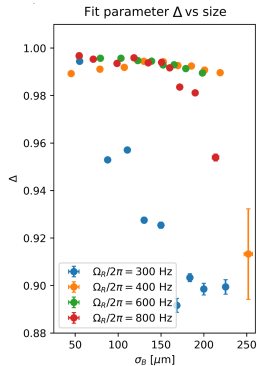
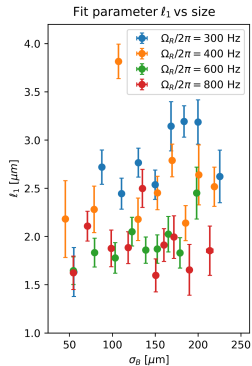
# ACF analysis in the outside region vs $\sigma_B$



- ▶ Shots clustered by increasing size (darker to lighter colors)
- ▶ Fit function:

$$\mathcal{A}_{\text{fit}}(x) = (1 - \Delta) \exp\left[-\frac{x}{2\ell_1}\right] + \Delta$$

# ACF analysis in the outside region vs $\sigma_B$ and $\Omega_R$



# Conclusions

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What did we learn?

- ▶ System shows **different properties** between inside and outside of the bubble
- ▶ Domain wall width **depends on the coupling** strength  $\Omega_R$
- ▶ Growth factor of the bubble size in time is **independent** of  $\Omega_R$
- ▶ In the bubble, periodic structures **disappear** with size increasing. They **appear**, instead, outside of the bubble.
- ▶ Length scale of information outside is related to the Rabi **healing length**

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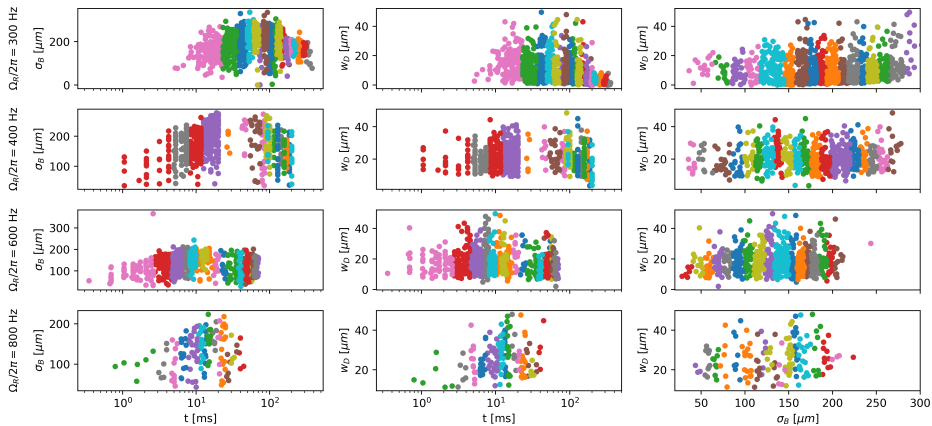
Future research:

- ▶ Analysis of the **density** channel
- ▶ Comparison with numerical GPE **simulations**
- ▶ Behavior at different **temperatures**

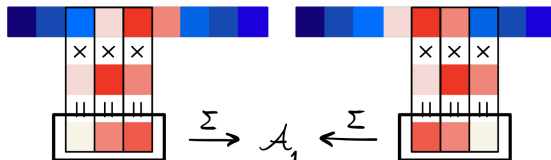
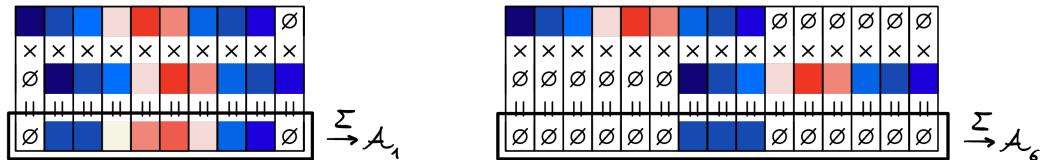


Thank you for your attention!

## Clustering



# Windowed ACF



# FFT and ACF averages in the inside region

